Theorem 4 (Informally)

To solve a recurrence

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + ... + c_k a_{n-k}$

Consider the characteristic polynomial (or equation)

$x^n - c_1 x^{n-1} - c_2 x^{n-2} - ... - c_k$

If r_i is a root with multiplicity of m_i

i.e $(x - r_i)$ is a factor,

then

$r_i^n, nr_n^n, n^2 r_n^n, ...n^{m_i - 1} r_i^n$

are all solutions to the recurrence,

The general solution is a linear combination of all such special solutions,

Solve for coefficients to match initial conditions.

Example

If the characteristic polynomial is

$(x-7)^5 (x+3)^3 \left(x + \sqrt{17} - 3\right)\left(x - \sqrt{17} - 3\right)$

Then our special solutions are:

$7^n, n7^n, n^2 7^n, n^3 7^n, n^4 7^n, (-3)^n, n(-3)^n, n^2(-3)^n, \left(-\sqrt{17} + 3\right)^n$ and $\left(\sqrt{17} + 3\right)^n$

A generic solution is thus

$c_n = \left(\alpha_{0,0} + \alpha_{0,1} n + a_{0,2} n^2 + \alpha_{0,3} n^3 + \alpha_{0,4} n^4\right) 7^n +$

$\alpha_{1,0} \alpha_{1,1} n + \alpha_{1,2} n^2)(-3)^n +$

$\beta\left(-\sqrt{17} + 3\right)^n + \gamma\left(\sqrt{17} + 3\right)^n$

There are 10 unknowns, and the recurrence is defined by 10 initial conditions

Note

The link between polynomials and recurrences works in both directions.

Suppose we want to compute $\left(\sqrt{17} - 3\right)^{80} = a + b\sqrt{17}$ for some a and b

We can use recurrences to compute this using exact integer arithmetic

Let $d_n = \left(\sqrt{17} - 3\right)^n$

Since this is of the form

$d_n = \alpha\left(\sqrt{17} - 3\right)^n + \beta\left(\sqrt{17} - 3\right)^n$

$\alpha = 1, \beta = 0$

It is a solution to a recurrence with characteristic equation

$\left(x - \sqrt{17} + 3\right)\left(x + \sqrt{17} + 3\right)) = x^2 + 6x - 8$

so

$d_n = -6d_{n-1} + 8d_{n-2}$

Now

$d_0 = 1,$

$d_1 = \sqrt{17} - 3,$

$d_2 = -6\left(\sqrt{17} - 3\right) + 8(1) = -6\sqrt{17} + 26$

$d_3 = -6\left(-6\sqrt{17} + 26\right) + 8\left(\sqrt{17} - 3\right) = 44\sqrt{17} - 180$

How is this connected to summation formulas?

Example

Suppose we want to compute

$a_n = \sum_{k=1}^{n} k2^k$

Example

$a_4 = \sum_{k=1}^{4} = k2^k$

$= 1 * 2^1 + 2 * 2^2 + 3 * 2^3 + 4 * 2^4$

$= 2 + 8 + 24 + 64$

$= 98$

Note

$a_n - a_{n-1} = \sum_{k=1}^{n} k2^k - \sum_{k=1}^{n-1} = k2^k$

$= n2^n + \left(\sum_{k=1}^{n-1} k2^n\right) - \left(\sum_{k=1}^{n-1} k2^k\right)$

$= n2^n$

So

$a_n = a_{n-1} + n2^n$ This is a recurrence, but it isnt homogenous

$a_n = a_{n-1}$ the recurrence part

$n2^n$ this is not zero

Let $b_n = a_n - a_{n-1}$

so $b_n = n2^n$

$b_n - 2b_{n-1} = n2^n - 2(n-1)2^{n-1}$

thus $= 2^n$

b_n also satisfies a recurrence , no homogenous

Let $c_n = b_n - 2b_{n-1}$

$c_n - 2c_{n-1} = 2^n - 2\left(2^{n-1}\right) = 0$

These new c_i's do satisfy a recurrence, By substituting, we actually get a recurrence for a

$c_n - 2c_{n-1} = 0$

so

$b_n - 2b_{n-1} - 2(b_{n-1} - 2b_{n-2}) = 0$

$b_n - 4b_{n-1} + 4b_{n-2} = 0$

But b_n is just $a_n - a_{n-1}$

so

$(a_n - a_{n-1}) - 4(a_{n-1} - a_{n-2}) + 4(a_{n-2} - a_{a-3}) = 0$

$a_n - 5a_{n-1} + 8a_{n-2} - 4a_{n-3} = 0$

$x^3 - 5x^2 + 8x - 4$ or $(x-1)(x-2)^2$

It must be

$a_n = \alpha l^n + \beta 2^n + \gamma n 2^n$ for some α, β, γ

since

$a_1 = 2, a_2 = 10, a_3 = 34$

we have

$2 = \alpha + 2\beta + 2\gamma$
$10 = \alpha + 4\beta + 8\gamma$
$34 = \alpha + 8\beta + 24\gamma$

3 equations in 3 unknowns

Take away

Non homogenous recurrences can be transformed (sometimes) into homogenous ones.

In general to solve

$a_n = c_1 a_{n-1} + c_2 a_{n-2} + ...c_k a_{n-k} + F(n)$

We find

a_n^h a homogenous solutions, (Ignore the $F(n)$)

and

a_n^p a particular solution

and glue them together. (†)

Theorem 5

The solution to † is of the form

$a_n = a_n^h + a_n^p$

- a_n^p There are parameters set to match $F(n)$
- a_n^h Parameters solved to match initial conditions

Theorem 6

If $F(n) = (b_t n^t + ...b_1 n + b_0)s^n$

then there is a particular solution of the form

$n^m(p_1n^t + ...p_1n + p_0)s^n$

where m is the multiplicity (possibly 0) of s as a root of the characteristic polynomial